

Initial Post

by [Abdulla Husain Salem Hadna Almessabi](#) - Wednesday, 6 August 2025, 10:54 AM

Number of replies: 4

Initial Post: Agent-Based Systems and Their Rise in Organisations

The organisational conditions are becoming dynamic, decentralised, and unpredictable, and so traditional centralised models of computing find it hard to provide flexibility, autonomy, and adaptability (Ding, Suiting, et al. 2023). ABS is a means of addressing these challenges, as it enables agent-agent interaction in a way that reflects real-world phenomena (Sayyed-Alikhani, Chica, & Mohammadi, 2021).

The ABS works best in settings that are defined by decentralisation in decision-making (Proselkov, Yaniv, et al. 2024). Software entities use information to make decisions and act without human interaction. They are programmed to solve a predetermined problem that necessitates instant action, like in the supply chain and customer service automation. Among them is the positive possibility of modelling complex behaviour within an organisation that gives insights into decision-making, resource allocation, and efficiency of operations (Aftabi, Navid, et al. 2025).

Agent-based systems boost efficiency, flexibility, and scalability in organisations. ABS is a way of optimising the distribution of resources, automating the decision-making process, and predicting future patterns by using data information (Mueller-Bloch, Christoph, et al. 2024).

Agent-based systems have become relevant in augmenting the operations of organisations as they provide a solution to the complexity and increased requirements faced by contemporary businesses. The advantages of independence, expansion, and flexibility make ABS one of the means to organisational success in the digital era.

References:

- Aftabi, N., Moradi, N., Mahroo, F. and Kianfar, F., 2025. SD-ABM-ISM: An integrated system dynamics and agent-based modeling framework for information security management in complex information systems with multi-actor threat dynamics. *Expert Systems with Applications*, 263, p.125681.
- Ding, S., Ward, H., Cucurachi, S. and Tukker, A., 2023. Revealing the hidden potentials of Internet of Things (IoT)-An integrated approach using agent-based modelling and system dynamics to assess sustainable supply chain performance. *Journal of Cleaner Production*, 421, p.138558.
- Mueller-Bloch, C., Andersen, J.V., Spasovski, J. and Hahn, J., 2024. Understanding decentralization of decision-making power in proof-of-stake blockchains: an agent-based simulation approach. *European journal of information systems*, 33(3), pp.267-286.
- Proselkov, Y., Zhang, J., Xu, L., Hofmann, E., Choi, T.Y., Rogers, D. and Brintrup, A., 2024. Financial ripple effect in complex adaptive supply networks: an agent-based model. *International Journal of Production Research*, 62(3), pp.823-845.
- Sayyed-Alikhani, A., Chica, M. and Mohammadi, A., 2021. An agent-based system for modeling users' acquisition and retention in startup apps. *Expert Systems with Applications*, 176, p.114861.